

Questions by topic: Ecological monitoring & new technologies - Set 1 (65 marks)

Answer all questions in the spaces provided.

0 1

A conservation team is planning to protect a population of an endangered antelope in a reserve.

0 1

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1 Explain why each of the following features of the population should be measured before a conservation strategy is decided.

[8 marks]

Population size

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Distribution

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Survival rate

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Age structure

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0 1

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2 Describe how the population size of the antelope could be estimated.

[2 marks]

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10

0 2 . 1

Explain how each of the following new technologies improves the validity of ecological research.

[8 marks]

Satellite / radio / GPS tracking

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DNA databases and eDNA

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Image recognition software

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Acoustic monitoring / sonograms

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0 2 . 2

Explain why new technologies allow the collection of 'more representative' data.

[2 marks]

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03 . 1

Which of the following would be most suitable for monitoring the movements of a whale over several months? Tick one box.

[1 mark]

- A Radio collar ☐
- B Acoustic / satellite tag ☐
- C Pitfall trap ☐
- D Geolocator tag recovered by recapture ☐

03 . 2

Explain why post-release monitoring is important in a captive breeding and release programme and describe **two** suitable methods.

[4 marks]

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5

04 . 1

Which technique would be most suitable for detecting whether great crested newts are present in a pond without catching them? Tick **one** box.

[1 mark]

- A Environmental DNA (eDNA) analysis ☐
- B Radio tracking ☐
- C Lincoln index (mark-release-recapture) ☐
- D Leg rings ☐

04 . 2

Explain why geolocator tags, rather than radio or GPS transmitters, are used to track the migrations of small birds.

[2 marks]

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0 4 . 3

Explain how DNA databases could help to control the illegal trade in elephant ivory.

[2 marks]

5

0 5 . 1

Complete Table 1 to show the environmental information that can be collected by satellite sensors detecting each type of radiation.

[3 marks]

Table 1

Radiation detected	Environmental information collected	
Reflected visible light		
Infrared emitted by the Earth's surface		
Reflected radar microwaves emitted by the satellite		

0 5 . 2

Explain why radar surveys are not interrupted by cloudy weather.

[1 mark]

05 . 3

Explain **two** advantages of satellite surveys over ground surveys for monitoring deforestation.

[2 marks]

05 . 4

Describe how motion-sensitive cameras and image databases could be used together to estimate the size of a tiger population.

[4 marks]

10

06 . 1

Explain why an electronic meter used to measure dissolved oxygen must be calibrated, and what should be done if the reading cannot be adjusted.

[2 marks]

06 . 2

Give **three** types of indirect evidence of the presence of an animal and the information each can provide.

[3 marks]

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2

3

0 6 . 3 Explain how auditory monitoring can be used to study bats. **[2 marks]**

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0 6 . 4 Explain **three** assumptions of the mark-release-recapture method that may not be met. **[3 marks]**

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07

Table 1 shows the number of adult great crested newts caught in bottle traps in a pond over five years, and the number of ponds in the area in which the species was detected using environmental DNA (eDNA).

Table 1

Year	Adults caught in traps	Ponds with newts detected by eDNA (out of 20)
2019	14	6
2020	22	9
2021	18	11
2022	31	13
2023	27	15

07

1

Calculate the percentage increase in the number of ponds where newts were detected between 2019 and 2023.

[2 marks]

Answer

%

07

2

Explain how eDNA detects the presence of newts without catching them.

[2 marks]

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07

3

Suggest why trap catches are a less reliable measure of population change than the eDNA survey.

[2 marks]

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Evaluate the contribution of new technologies to ecological monitoring and wildlife conservation.

[9 marks]

Do not write outside the box

Extra space

15

END OF QUESTIONS